



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Production Engg.
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: SY B.Tech.

Course: Engineering Metallurgy

Course Code: PCC-PE204

Date: 14.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

REG.NO.....

Note:

- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	What is cast iron? State the difference between steel and cast iron.	05	CO1	BT1
	b	What is white cast iron? Describe its structure, properties, and common applications.	05	CO2	BT1
Q.2	a	Describe the normalizing process and explain how it improves the mechanical properties of steel	05	CO3	BT2
	b	Explain the heat treatment process performed after carburizing and its purpose.	05	CO3	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Identify the methods of manufacturing of iron and steel.
CO2	Interpret significance of Iron-Iron carbide equilibrium diagram, time Temperature transformation (TTT) diagrams and powder metallurgy.
CO3	Classify the various metals alloys and composite materials applications and heat treatment processes.
CO4	Evaluate a metal alloy or material for a practical case.
CO5	Select the suitable non-destructive testing method for inspecting components.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: SY B.Tech.	Subject: Manufacturing Processes	Subject Code: PCC-PE203
Date: 14.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. Q1 & Q2 are compulsory.

Q.N.		Marks	CO	BT
Q.1	Solve any one from Q1 (a) and (b).			
a	Explain in detail with sketches the procedure and significance of following (any two) 1. Tension testing of welds. 2. Submerged arc welding. 3. Ultrasonic examination of weld.	10	CO2	BT3
b	Write down the description and causes for the following defects in welding along with neat sketches: i. Incomplete penetration ii. Undercutting iii. Porosity iv. Overlap. v. Distortion.	10	CO2	BT1
Q.2	Solve any two from Q2 (a), (b) and (c).			
a	Explain in detail the any five patterns with labelled sketches.	05	CO1	BT1
b	Explain in detail various types of pattern allowances and their importance in casting.	05	CO1	BT2
c	Explain in detail various types of pattern materials and their use in casting.	05	CO1	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1:	To Identify casting processes, working principles and applications and list various defects in metal casting.
CO2:	To classify the basic joining processes and demonstrate welding, brazing and soldering principles.



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded
Department of Production Engineering
In-Semester Examination II
Academic Year: 2025-26 (Sem-I)

Class: SY B.Tech.

Subject: Thermal Engineering I

Subject Code: PCC-PE202

Date: 15.10.2025

Time: 02.30 pm to 03.15 pm

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Figures to right indicate full marks.
- iii. The symbols carry usual meaning.
- iv. Solve any 4 questions from the following.

Q. N.	M	CO	BT
Q.1 What is Boiler? Write in brief the classification of boilers.	05	CO1	BT1
Q.2 Explain in brief the parameters used to estimate the performance of boilers.	05	CO1	BT2
Q.3 Compare boiler mountings and boiler accessories.	05	CO1	BT2
Q.4 What is steam condenser. Enlist the functions of a condenser.	05	CO4	BT2
Q.5 Explain in brief the different elements of condensation plant.	05	CO4	BT2
Q.6 Explain the function of steam nozzle. Explain the working of steam nozzle.	05	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

- CO 1 Describe fundamentals of thermodynamics, thermodynamic devices and their applications.
- CO 2 Spectacle the knowledge in the field of steam engineering.
- CO 3 Evaluate the performance of I.C. Engine.
- CO 4 Identify the thermodynamics devices required in different thermal engineering applications

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded-431606.	
	In Semester Examination-II (Semester-I) 2025-26	
	S.Y. B. Tech.	Department: Production Engineering
	Machine Drawing and CADD	PCC-PE205
Date: 11/10/2025	Time: 14:30 - 15:15	Max. Marks: 20

Note:

- Attempt all questions
- Assume suitable data if necessary.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Questions	Marks	CO	BT
Q.1	Enlist and draw different types of forms and proportions of rivet-heads.	12	CO4	BT1,6
Q.2	Write any 2 commands in AutoCAD for each of the following: - Geometry creation - Editing - Viewing - Printing	08	CO5	BT2,3

COs (Course Outcomes):

- CO 1: To acquire knowledge of various ISO standards used in machine drawing and apply those.
- CO 2: To read and interpret the conventions displayed on a product drawing.
- CO 3: To use the important entities like Limits, Fits, Tolerances and Surface Finish, towards their use in the drawing.
- CO 4: To identify and draw various machine parts.
- CO 5: To draw/model parts and assemblies using CAD software.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create.



DT: 10/10/25
14.30-15.15

SGGS INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED – 431 606

Department of Production Engineering

Academic Year 2025-26

ISE-2 Examination Class : Class: S.Y. Production

Subject: CASTING AND WELDING (PCC-PE-203)

Max Marks: 20

Note: Solve any TWO questions from the following.

***CO: Course Outcomes., BT-blooms taxonomy**

Q.1	With help of neat labeled sketch explain the principle, working and advantages of submerged arc welding process.	10	CO3	BT2
Q.2	Explain with neat sketch the principle and working of radiographic examination used for testing of welding with advantages and limitations	10	CO1	BT3
Q.3	Draw and explain with a neat sketch to represent cope, drag, and gating system for a sand mold casting. Also label different elements in the sketch.	10	CO1	BT2

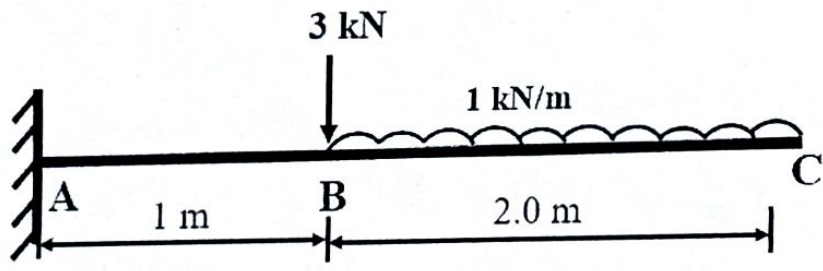


Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: SY B.Tech.	Course: Strength of Materials	Course Code: PCC-PE201
Date: 13.10.2025	Time: 02.30 pm to 03.15 pm	Max. Marks: 20

Note:

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- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Calculate shear force and bending moments at various cross sections of the cantilever beam as shown below. Draw their variations along the length for the beam. 	12	CO3	BT3
Q.2	A steel plate is bent into a circular arc of radius 15 meters. If the plate section is 100 mm wide and 10 mm thick, find the maximum stress induced and the bending moment which can produce this stress. Take $E = 2 \times 10^5 \text{ N/mm}^2$.	08	CO4	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

- CO 1 To differentiate the mechanical behaviour of engineering materials.
- CO 2 To apply the theories of elastic failure.
- CO 3 To determine the shear force and bending moment distribution of the beam and locate the point of contraflexure.
- CO 4 To calculate stresses in beams for various boundary conditions.
- CO 5 To determine the slope and deflection of the beam for various boundary conditions.